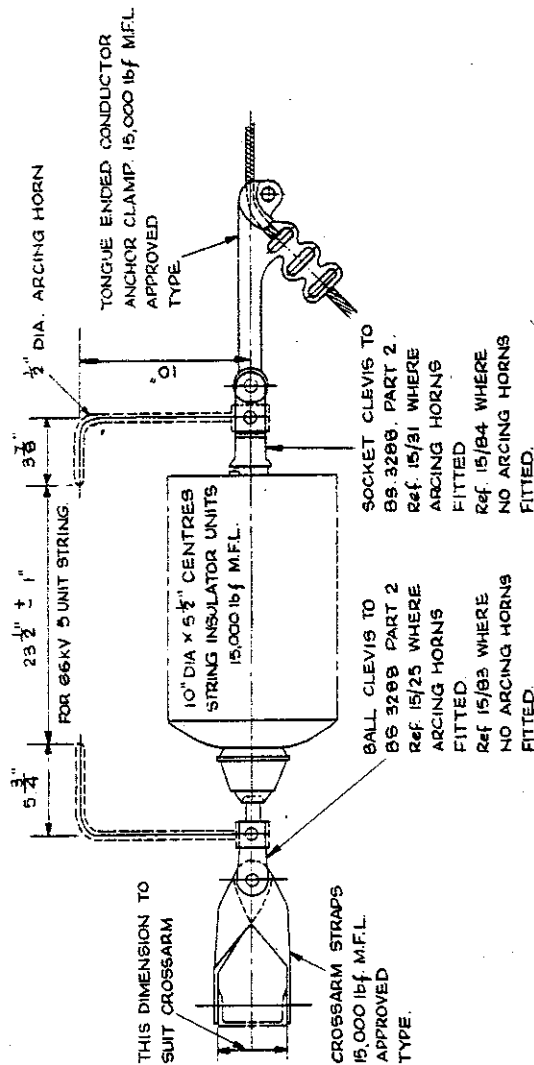


6. ALL DUPLICATION INSULATOR SETS SHOULD BE ELIMINATED.



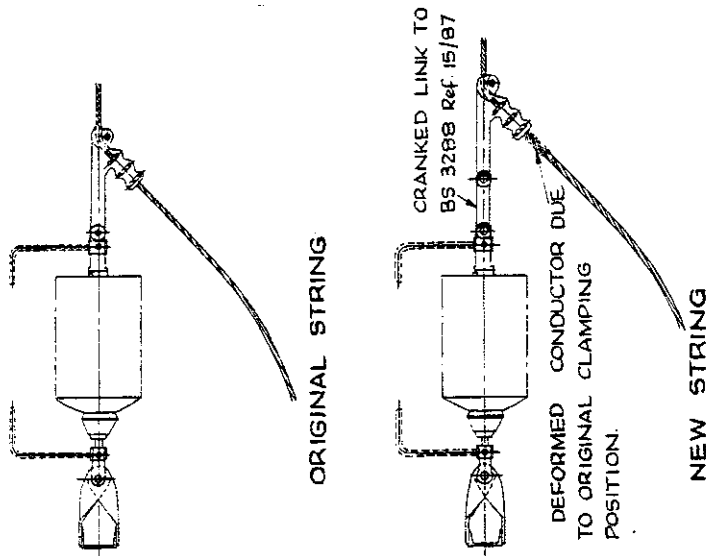
ARRANGEMENT

NOTES

1. THIS DRAWING APPLIES ONLY TO LINES WITH CHANNEL TYPE CROSSARMS LINES WITH ANGLE TYPE CROSSARMS SHOULD BE MAINTAINED IN ACCORDANCE WITH DRG. NO Y003X 2204 (28000 lbf MFL) OR Y003X 2207 (15000 lbf MFL)
2. ARGING HORNS
 - 11KV LINES
 - 33KV LINES
 - 66KV UNEARTHED SUPPORTS
 - 66KV EARTHED SUPPORTS
 - 3. STRING INSULATOR UNITS REQUIRED
 - a. ALL 11KV SUPPORTS
 - 2 UNITS
 - b. 33KV UNEARTHED SUPPORTS
 - 3 UNITS
 - c. 33KV EARTHED SUPPORTS
 - 4 UNITS
 - d. ALL 66KV SUPPORTS
 - 5 UNITS

4. CONDUCTOR ANCHOR CLAMPS SHOULD HAVE HAD THE ROUTINE TESTS SPECIFIED IN B.S. 3288 PART 1. APPLIED BETWEEN THE EYE AND TONGUE OR CLEVIS OF THE CLAMP.
5. REPLACEMENT STRINGS SHOULD BE EQUAL TO OR SLIGHTLY LONGER THAN THE ORIGINAL STRINGS SO THAT THE CONDUCTOR IS CLAMPED IN THE SAME PLACE, OR THE FORMER CLAMPING POINT IS MOVED INTO THE BOW. A CRANKED LINK MAY BE INTRODUCED INTO THE STRING FOR THIS PURPOSE.

REPLACEMENT STRINGS (SEE NOTE 5)



drawn	E.W.
date	J.A.
checked	U
date	28.3.69
scale	1/2" = 4' TO 1 FT.

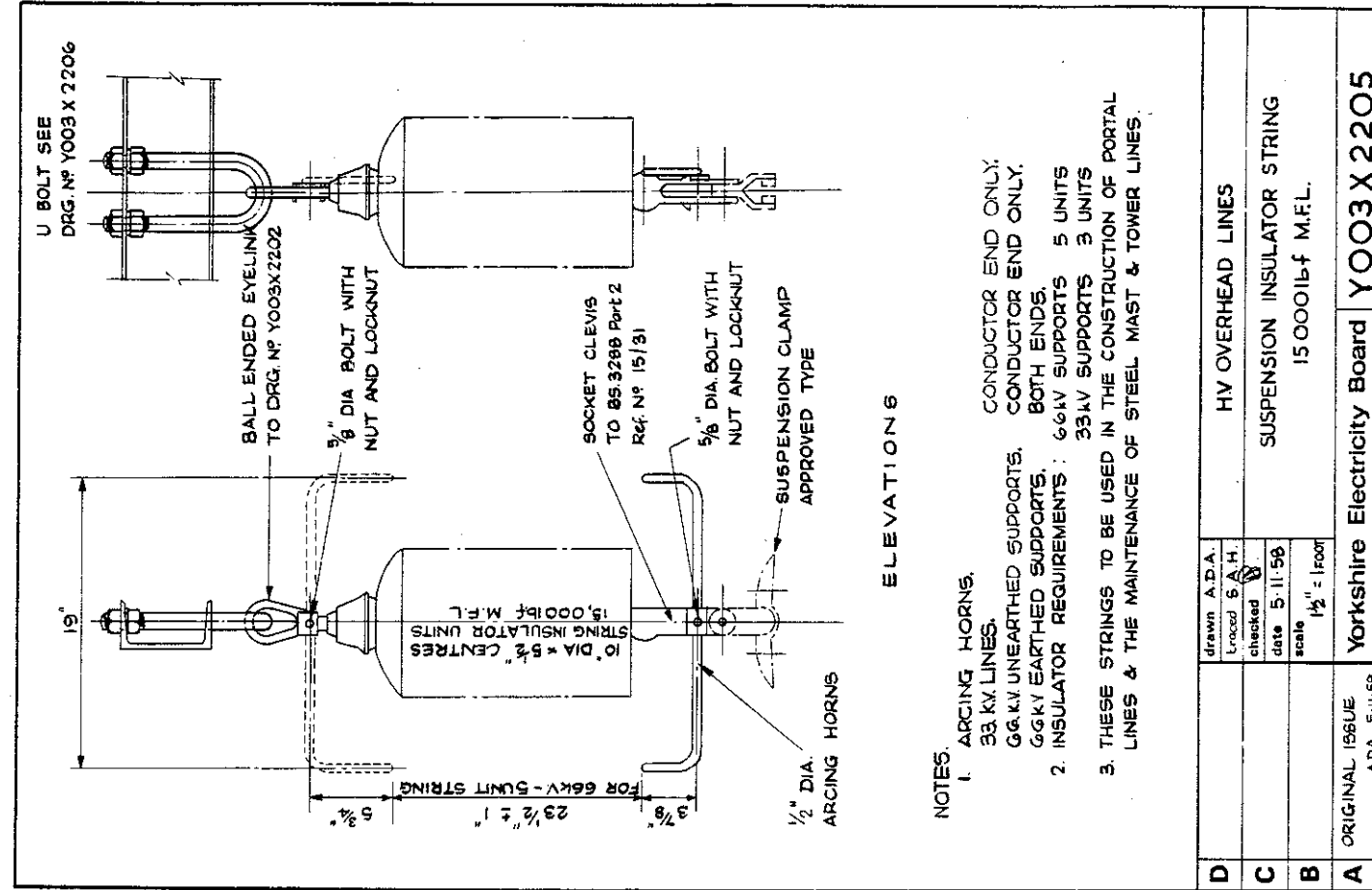
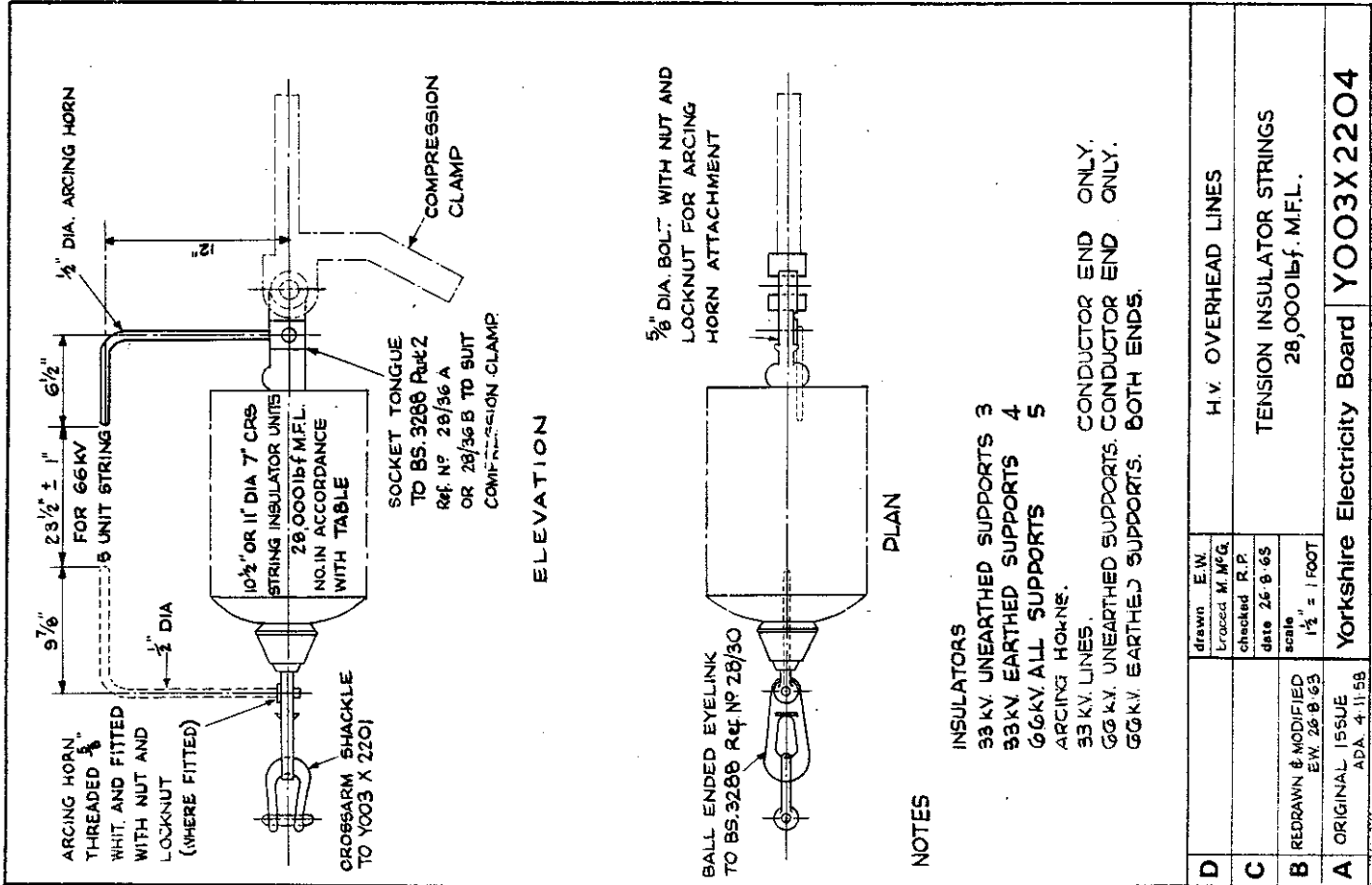
HV OVERHEAD LINES

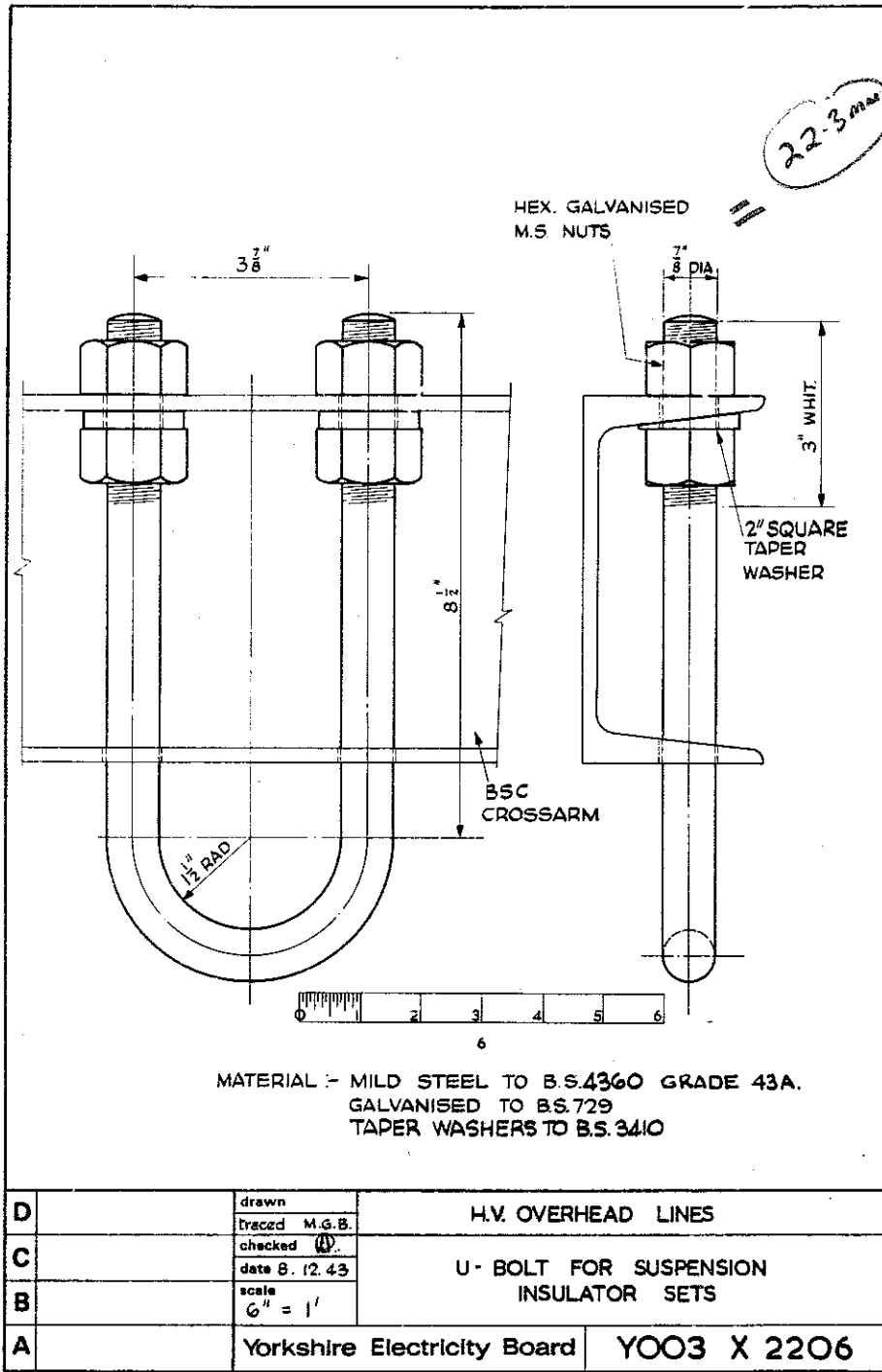
TENSION INSULATOR STRINGS 15000 lbf MFL.
(FOR MAINTENANCE OF LINES WITH B.S.C. CROSSARMS & COPPER CONDUCTORS)

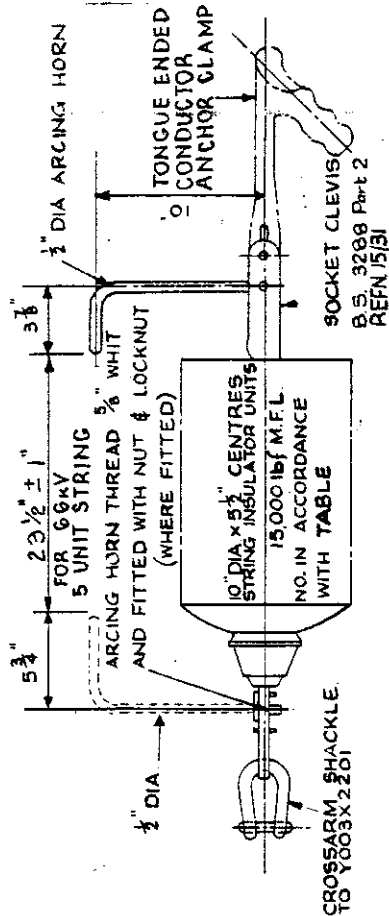
A NOTE 3 & 4 DELETED B NOTE 5 REVISED C

Yorkshire Electricity Board

Y003X 2203







ELEVATION



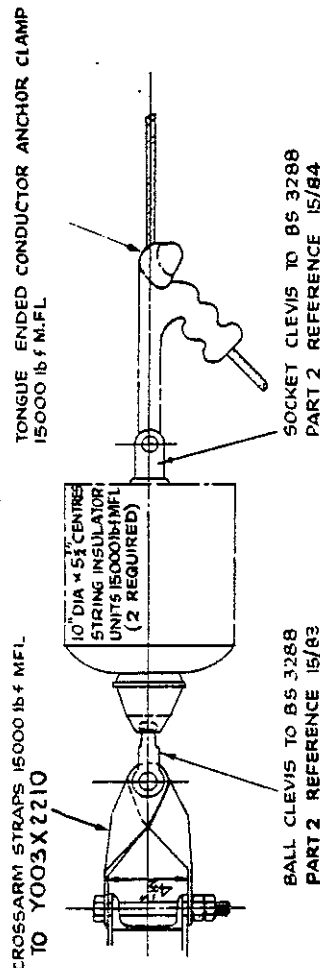
PLAN

NOTES
INSULATORS
33KV UNEARTHED SUPPORTS 3
33KV EARTHED SUPPORTS 4
66KV ALL SUPPORTS 5

ARCING HORNS

33KV LINES. CONDUCTOR END ONLY.
66KV UNEARTHED SUPPORTS. CONDUCTOR END ONLY.
66KV EARTHED SUPPORTS. BOTH ENDS.

D	66KV EARTH SUPPORT WAS 6 INSULATORS	drawn E.W. 15.10.67	traced J.A.	33 KV & 66 KV OVERHEAD LINES
C	ALTERNATIVE REMOVED	checked R.P. 15.10.63	date 28.8.63	TENSION INSULATOR STRINGS
B	REDRAWN & MODIFIED	scale 1 1/2" = 1'-0"	scale 1 1/2" = 1'	BSUA & BSEA CROSSARMS
A	ORIGINAL ISSUE	Yorkshire Electricity Board Y003X2207		



D	drawn D.C. traced C.D.M.	OVERHEAD LINES SPEC CE/C/31
C	checked 21.5.69	TENSION INSULATOR STRINGS
B	scale 1 1/2" = 1'	(15000 lbf M.F.L.)
A	ALTERNATIVE DELETED	Yorkshire Electricity Board Y003X2208



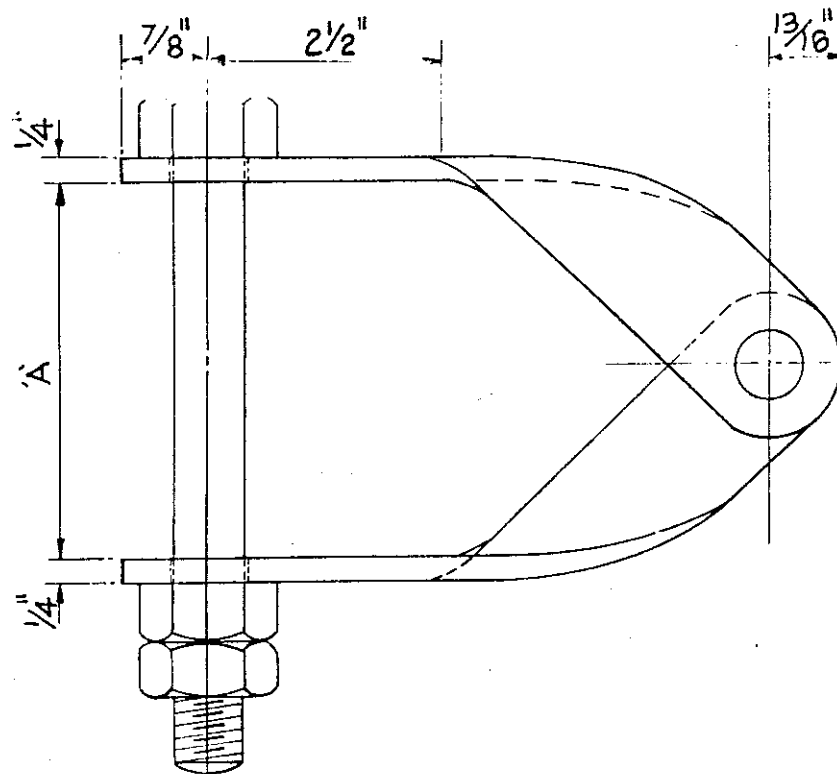
3	33 kV	UNEARTHED SUPPORTS
4 <th>33 kV</th> <th>EARTH SUPPORTS</th>	33 kV	EARTH SUPPORTS
5 <th>66 kV</th> <th>ALL SUPPORTS</th>	66 kV	ALL SUPPORTS

2. ARcing HORNS
33 KV LINES CONDUCTOR END ONLY
66 KV UNEARTHED SUPPORTS CONDUCTOR END ONLY
66 KV EARTHED SUPPORTS BOTH ENDS

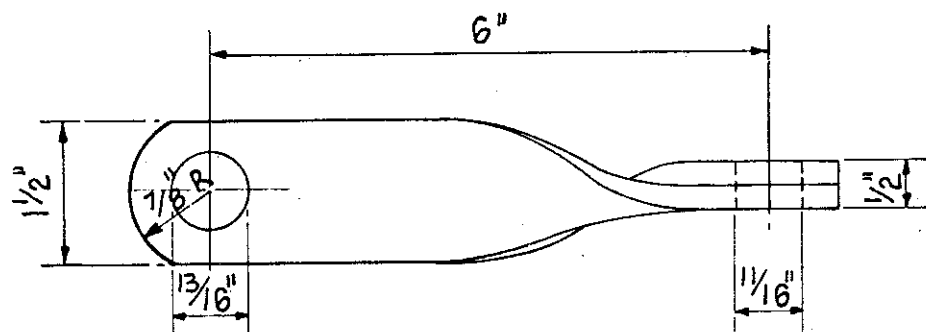
- ### 3 FOR MAINTENANCE OF LINES WITH BSC CROSSARMS AND ACSR CONDUCTORS

A SOCKET TONGUE REF 28/36B WAS 28/36A.	B	C	D	Yorkshire Electricity Board	Y003X 2209
				33 kV & 66 kV OVERHEAD LINES	
				drawn N.F. traced G.D.M.	
				checked <u>3/4/80</u> date 9. 10. 70	
				scale $\frac{1}{8}'' = 1'$	
TENSION INSULATOR STRING 28000lb MFL FOR LINES WITH BSC CROSSARMS					

[illegible]



ELEVATION



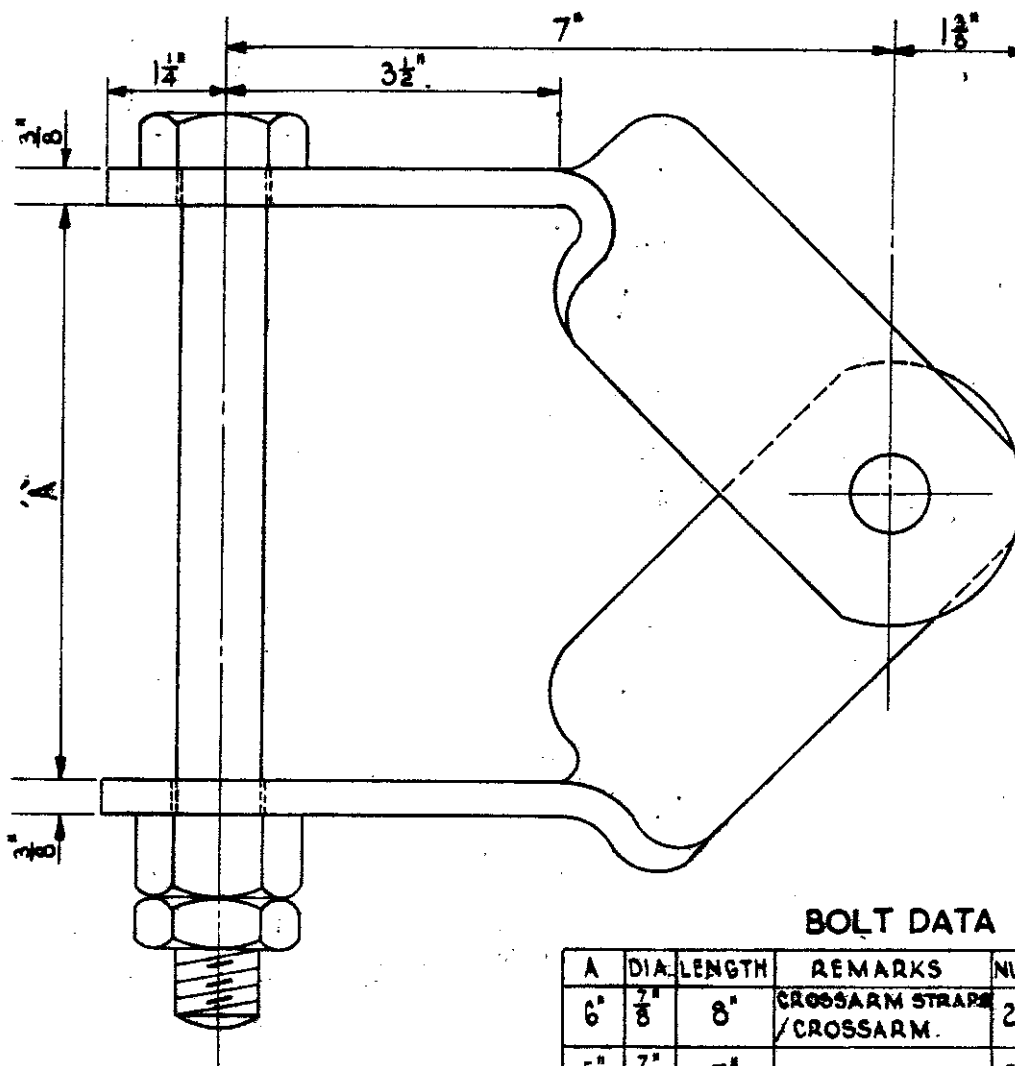
PLAN

BOLT DATA

A	DIA	LENGTH	REMARKS	NUT	STR'D
5 1/2"	3/4"	8"	CROSSARM STRAPS/CR ARM	2	2 1/2"
4 1/2"	3/4"	6 1/2"	"	2	2"

BOLTS TO BS 916 THREADED BSW SUPPLIED WITH STRAP MATERIAL - MILD STEEL GALVANISED TO BS 729

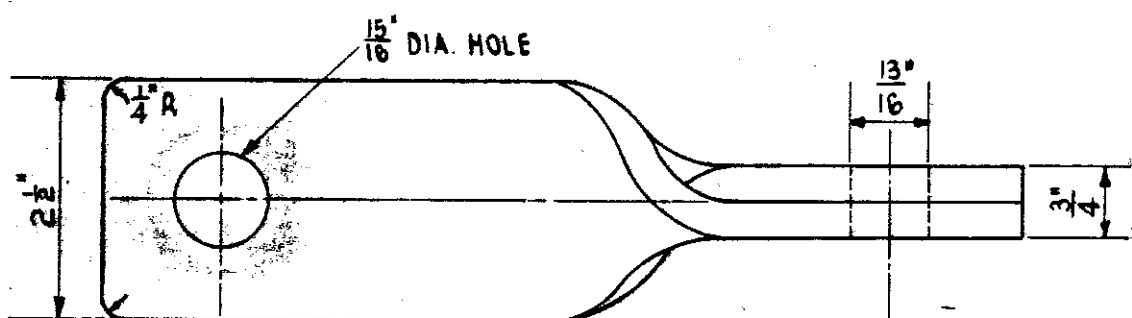
D		drawn N.F.	HV. OVERHEAD LINES
		traced RS-S	
C	7/8" RADIUS WAS 1" SQUARE RS-S-24-57	checked JKS	
		date 24-9-70	
B	DIMENSION A ALTERED RS-S-5-11-70	scale 1/2 F.S.	CROSSARM STRAP 15000lb M.F.L.
A	END OF STRAP MODIFIED	Yorkshire Electricity Board	Y003X2210



ELEVATION

BOLT DATA

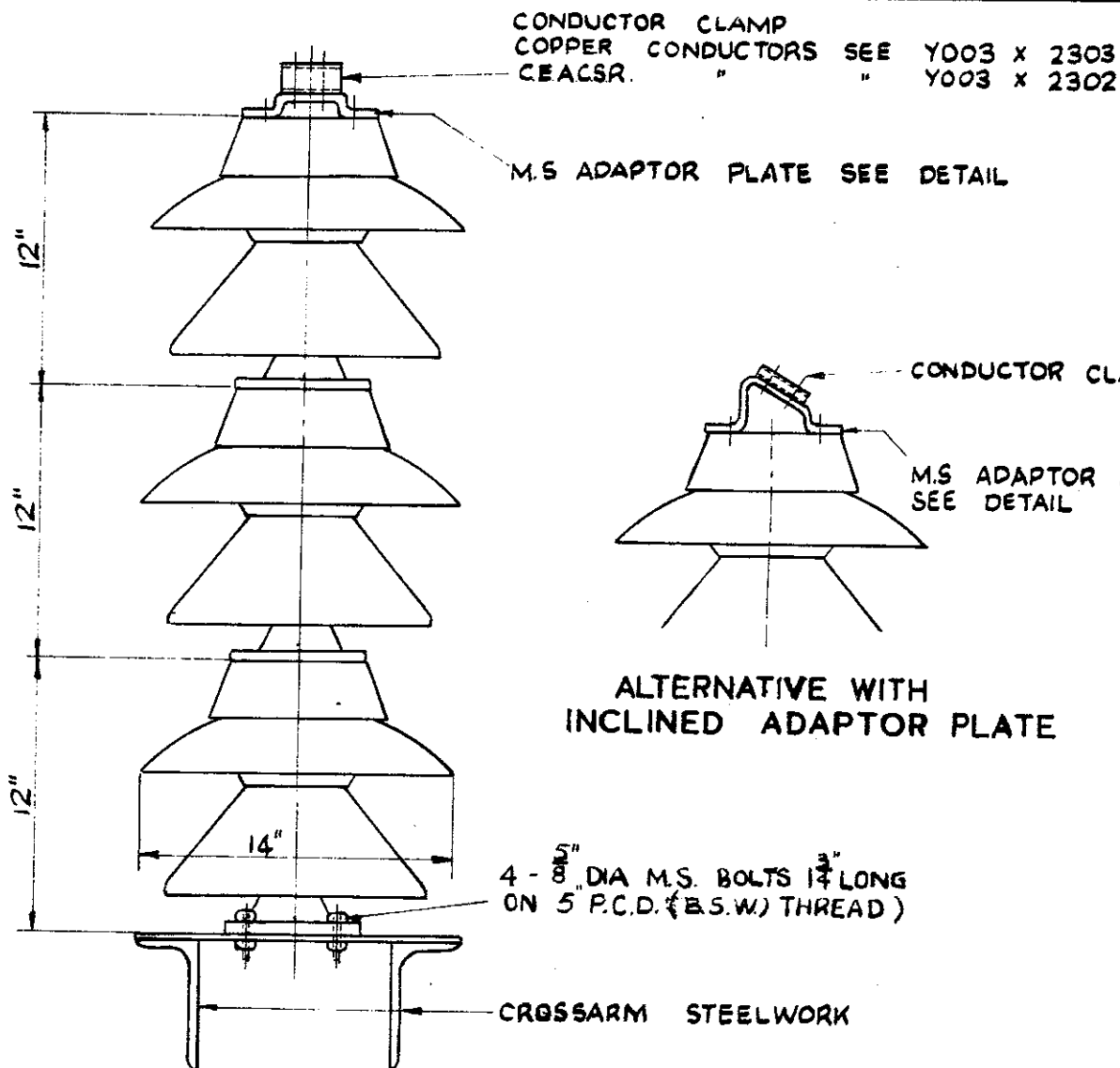
A	DIA.	LENGTH	REMARKS	NUTS	THR'D
6"	7/8"	8"	CROSSARM STRAP / CROSSARM.	2	2 1/2"
5"	7/8"	7"	"	2	2 1/2"



PLAN

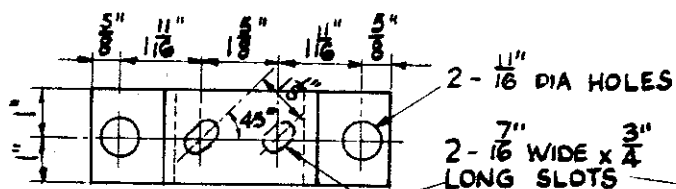
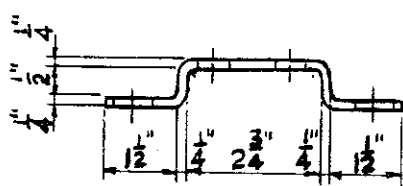
BOLT TO BS 916 THREADED BSW SUPPLIED WITH STRAP
MATERIAL :- MILD STEEL GALVANISED TO BS 729

D	drawn	N. F.	HV. OVERHEAD LINES
	traced	R. S. S.	
	checked	JLS	
	date	25-9-70	
C	scale	1/2 FULL SIZE	CROSSARM STRAP 28,000 lb f. M F L.
B			
A			
Yorkshire Electricity Board			Y003X2211

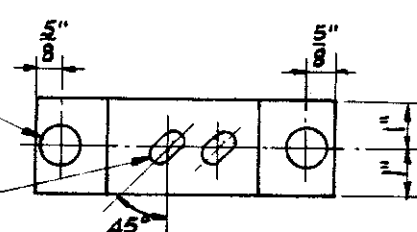
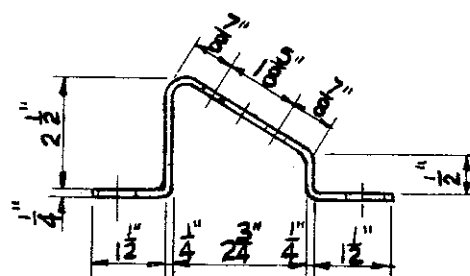


ALTERNATIVE WITH
INCLINED ADAPTOR PLATE

POST INSULATOR
3 - 12" HIGH x 14" DIA UNITS
SCALE: - 1¹/₂" = 1'



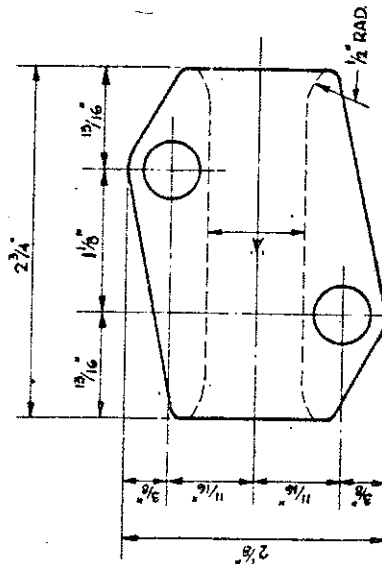
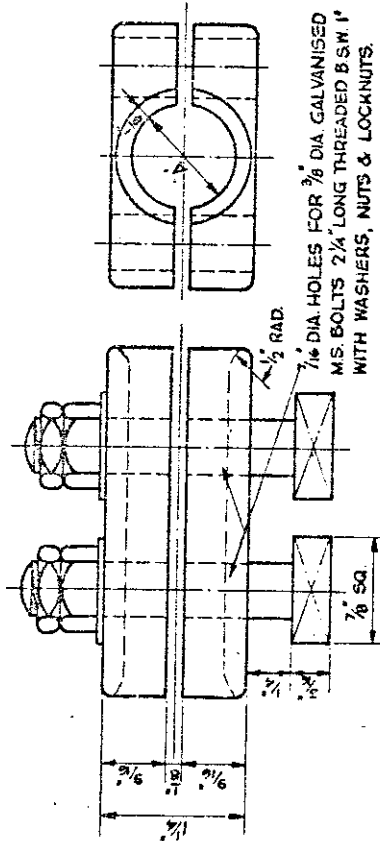
M.S. ADAPTOR PLATE
SCALE: - 1/4 FULL SIZE



M.S. INCLINED ADAPTOR PLATE
SCALE: - 1/4 FULL SIZE

NOTE: ALL STEELWORK GALVANISED TO B.S. 729

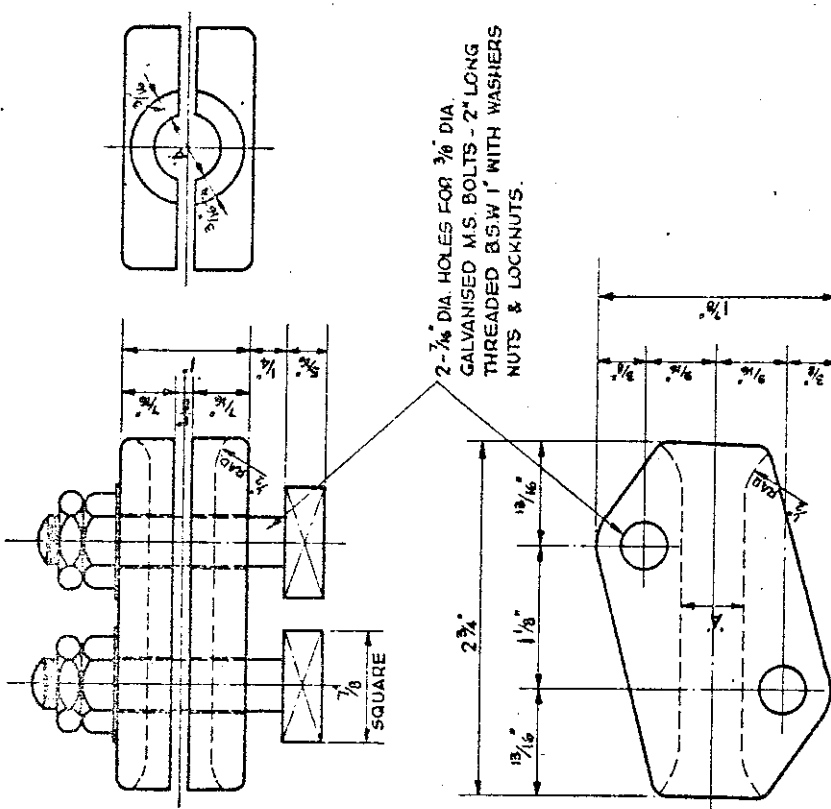
D		drawn	H V OVERHEAD LINES PILOT INSULATOR ARRANGEMENT FOR 66 KV EARTHED SUPPORTS
		traced K.E.W.	
C		checked <i>SL</i>	
		date 31.2.69	
B		scale	
A		Yorkshire Electricity Board	Y003 X 2301



DIM. A.	CONDUCTOR SIZE	Nº REQD.
23/32"	0.15 sq. ins. 37/102"	
25/32"	0.175 sq. ins. 37/110"	

MATERIAL :- ALUMINIUM ALLOY

D	drawn A.J.	66KV. OVERHEAD LINES
C	checked M.G.B.	
B	checked R.P.	
A	date 5.11.57	
	scale FULL SIZE	
	FILE BLOCK ALTERED TO 1/2" DIA POST INSULATOR CLAMP FOR ACSR CONDUCTORS	
	RE-TRACED	
	M.G.B. 13.1.55	
	Yorkshire Electricity Board	Y003X2302



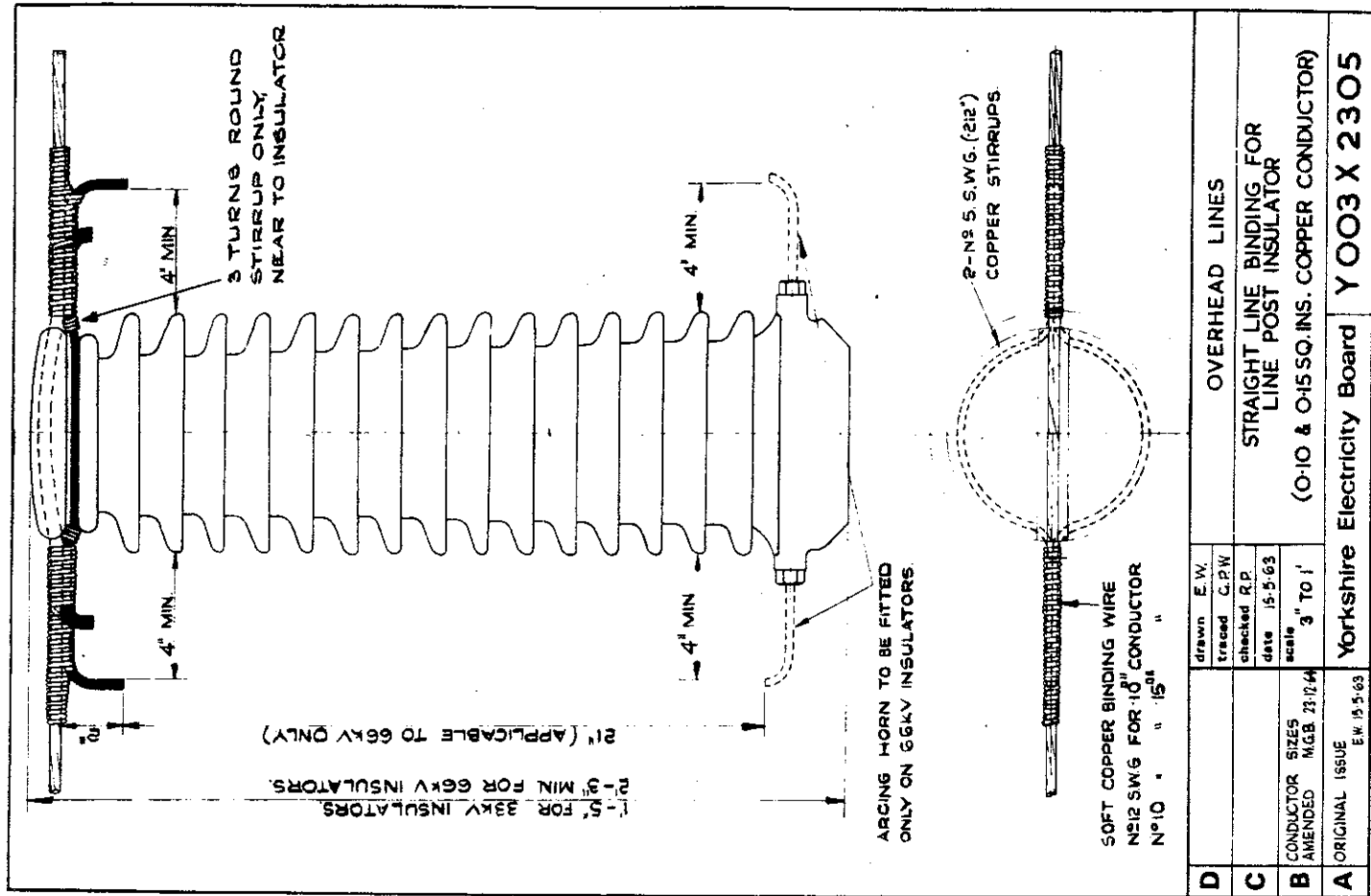
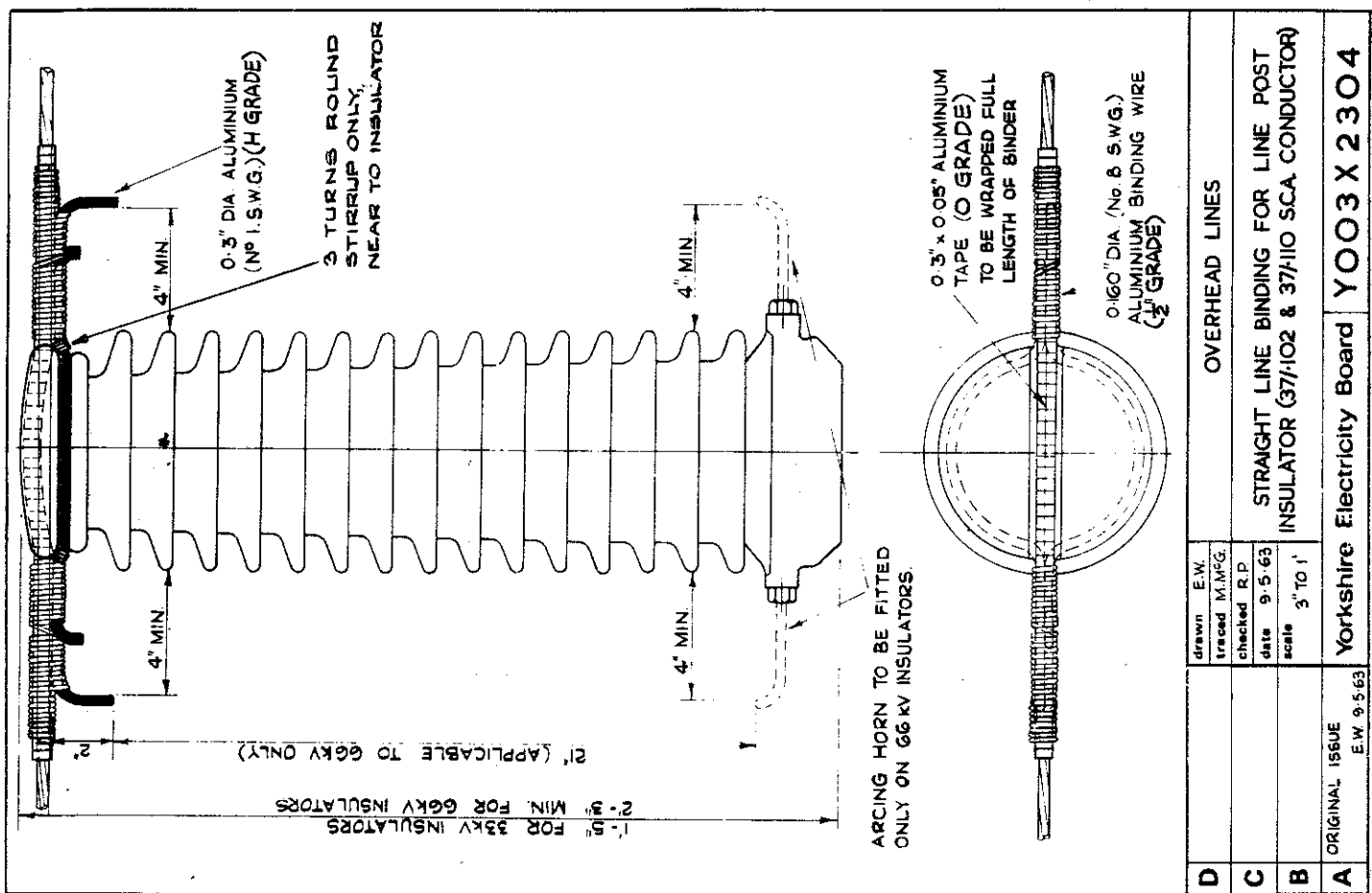
COMPOSITION OF
BRONZE

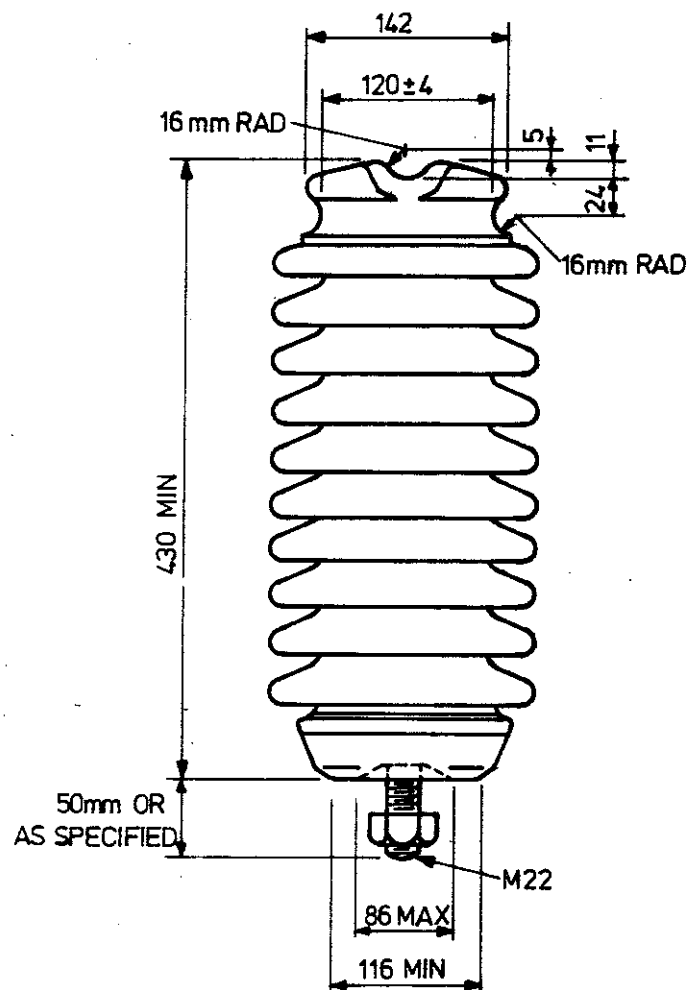
MIXTURE OF METALS	%
COPPER	87 1/2
TIN	10
ZINC	2
LEAD	1/2
TOTAL	100

DIM. A.	CONDUCTOR SIZE	Nº REQD.
23/32"	0.10 sq. ins. 7/136"	
1/4"	0.15 sq. ins. 7/166"	
19/32"	0.20 sq. ins. 7/193"	

MATERIAL :- BRONZE

D	drawn A.D.A.	66KV. OVERHEAD LINES
C	checked M.G.B.	
B	checked R.P.	
A	date 13.10.58	
	scale FULL SIZE	
	FILE BLOCK ALTERED TO 1/2" DIA POST INSULATOR CLAMP FOR COPPER CONDUCTORS	
	RE-TRACED	
	M.G.B. 13.1.55	
	Yorkshire Electricity Board	Y003X2303





TOTAL CREEPAGE DISTANCE = 900 MIN

- 1 DIMENSIONS AND CHARACTERISTICS TO COMPLY WITH ESI STANDARD 43-93 AND BS 137 FOR A NOMINAL SYSTEM VOLTAGE (3 PHASE) OF 33 kV
MINIMUM TRANSVERSE FAILING LOAD TO BE 10 kN
- 2 TOLERANCES EXCEPT WHERE SPECIFIED ON DRG ARE TO COMPLY WITH BS 137
- 3 INSULATOR SUPPLIED COMPLETE WITH GALVD. MAL. C.I. BASE AND FITTED WITH GALVD. M22 STUD 50mm IN LENGTH BELOW BASE AND ONE NUT TO COMPLY WITH BS 3288 PART 2
- 4 INSULATOR TO BE INDELIBLY MARKED IN ACCORDANCE WITH BS 137
- 5 IN LOCALITIES NOT SUBJECT TO HEAVY SALT OR INDUSTRIAL DEPOSITS, THE PIN INSULATOR TO DRG Y003X2106 MAY BE USED
- 6 PORCELAIN TO BE GLAZED BROWN



D		drawn KLS	33 kV LINE POST INSULATOR
		traced JMW	
C		checked	FOR USE IN LOCALITIES SUBJECT TO HEAVY SALT OR INDUSTRIAL DEPOSITS
		date 26.1.66	
B		original scale 1:5	ESI DRAWING 439303
A	REVISED IN LINE WITH ESI STANDARD JMW 5.4.77	Yorkshire Electricity Board	Y003 X 2306